

What is DNS?

DNS = Domain Name System

It converts **domain names to IP addresses**

Computers communicate using **IP addresses**

DNS is called the **phonebook of the Internet**

Linux DNS configuration file:

/etc/resolv.conf

Item	Value	Meaning
Server	DNS	Converts domain names to IP addresses
Software	BIND	DNS server software
Package	bind	Installs the DNS server
Service	named	Runs the DNS server
Main Config File	/etc/named.conf	Main DNS configuration
Zone Files	/var/named/	Stores forward and reverse zone files
Port	53	DNS communication
Protocol	UDP/TCP	UDP for queries, TCP for zone transfers
Start Service	systemctl start named	Starts DNS service
Enable Service	systemctl enable named	Starts DNS automatically after boot
Check Status	systemctl status named	Checks if DNS is running

#yum install packages

```
[root@server ~]# yum install bind bind-utils
CentOS Stream 9 - BaseOS                                4.3
CentOS Stream 9 - AppStream                             3.2
CentOS Stream 9 - Extras packages                       7.5
Package bind-utils-32:9.16.23-40.el9.x86_64 is already installed.
Dependencies resolved.
=====
Package                        Architecture      Version                               Repository
=====
Installing:
bind                          x86_64            32:9.16.23-43.el9                   appstream
Upgrading:
bind-libs                    x86_64            32:9.16.23-43.el9                   appstream
bind-license                 noarch            32:9.16.23-43.el9                   appstream
bind-utils                   x86_64            32:9.16.23-43.el9                   appstream
Installing dependencies:
bind-dnssec-doc              noarch            32:9.16.23-43.el9                   appstream
python3-bind                 noarch            32:9.16.23-43.el9                   appstream
python3-ply                  noarch            3.11-14.el9                        baseos
Installing weak dependencies:
bind-dnssec-utils            x86_64            32:9.16.23-43.el9                   appstream
```

Check the ping or not ping in server to client

```
valid_ttl forever preferred_ttl forever
[root@server ~]# ping 192.168.15.128
PING 192.168.15.128 (192.168.15.128) 56(84) bytes of data.
64 bytes from 192.168.15.128: icmp_seq=1 ttl=64 time=11.3 ms
64 bytes from 192.168.15.128: icmp_seq=2 ttl=64 time=0.795 ms
64 bytes from 192.168.15.128: icmp_seq=3 ttl=64 time=0.712 ms
64 bytes from 192.168.15.128: icmp_seq=4 ttl=64 time=0.687 ms
64 bytes from 192.168.15.128: icmp_seq=5 ttl=64 time=1.15 ms
^C
--- 192.168.15.128 ping statistics ---
```

Client to server

```
[root@client ~]# ping 192.168.15.130
PING 192.168.15.130 (192.168.15.130) 56(84) bytes of data.
64 bytes from 192.168.15.130: icmp_seq=1 ttl=64 time=8.85 ms
64 bytes from 192.168.15.130: icmp_seq=2 ttl=64 time=4.58 ms
64 bytes from 192.168.15.130: icmp_seq=3 ttl=64 time=1.22 ms
64 bytes from 192.168.15.130: icmp_seq=4 ttl=64 time=1.27 ms
64 bytes from 192.168.15.130: icmp_seq=5 ttl=64 time=1.07 ms
64 bytes from 192.168.15.130: icmp_seq=6 ttl=64 time=1.25 ms
64 bytes from 192.168.15.130: icmp_seq=7 ttl=64 time=1.43 ms
64 bytes from 192.168.15.130: icmp_seq=8 ttl=64 time=1.31 ms
```

Step2

Configuration File

Main configuration file

```
#vim /etc/named.conf
```

```

attacks. Implementing DNSSEC within your network would greatly
reduce such attack surface
*/
recursion yes;

dnsssec-validation yes;

managed-keys-directory "/var/named/dynamic";
geoip-directory "/usr/share/GeoIP";

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
Already at oldest change

```

Then dd – delete the logging below the line all and save exit

```
pid-file "/run/named/named.pid";  
session-keyfile "/run/named/session.key";  
  
/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */  
include "etc/crypto-policies/back-ends/bind.config";  
  
};
```

:wq!

Step 3

1.Copy the # ip a

```
[root@server ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
    link/ether 00:0c:29:28:c6:cc brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.15.130/24 brd 192.168.15.255 scope global dynamic noprefixroute ens33
        valid_lft 1648sec preferred_lft 1648sec
    inet6 fe80::20c:29ff:fe28:c6cc/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

2.Go to the add the after the port 53 { 127.0.0.1; 192.168.15.130; } and add the {localhost: any; }

```
[root@server ~]# vim /etc/named.conf
```

```
[root@server named]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; 192.168.15.130; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { localhost; any; };
```

```
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; 192.168.15.130; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { localhost; any; };
```

Step 3

```
#vim /etc/named.rfc 1912.zones
```

Copy the four zones

Then I go to `#vim /etc/named.conf` I pastes the zones

```
zone "shelldecoders.in" IN {
    type master;
    file "db_forward";
    allow-update { none; };
};
```

```
zone "localhost" IN {  
    type master;  
    file "named.localhost";  
    allow-update { none; };  
};  
  
zone "1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ip6.arpa" IN {  
    type master;  
    file "named.loopback";  
    allow-update { none; };  
};  
  
zone "1.0.0.127.in-addr.arpa" IN {  
    type master;  
    file "named.loopback";  
    allow-update { none; };  
};
```

Then I change the 1. localhost into shelldecoders.in 2.file name db_forward

```
};  
  
zone "shelldecoders.in" IN {  
    type master;  
    file "db_forward";  
    allow-update { none; };  
};  
  
zone "localhost" IN {  
    type master;  
    file "named.localhost";  
    allow-update { none; };  
};  
  
zone "1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ip6.arpa" IN {  
    type master;  
    file "named.loopback";  
    allow-update { none; };  
};
```

Save and exit

I ckeck the #cat /etc/named.conf

```
};  
  
zone "shelldecoders.in" IN {  
    type master;  
    file "db_forward";  
    allow-update { none; };  
};
```

Step4

I go #cd /var/named

```
[root@server ~]# cd /var/named  
[root@server named]# ls  
data dynamic named.ca named.empty named.localhost named.loopback slaves  
[root@server named]# ls -l  
total 16  
drwxrwx---. 2 named named    6 Apr 17 16:32 data  
drwxrwx---. 2 named named    6 Apr 17 16:32 dynamic  
-rw-r-----. 1 root  named 2112 Apr 17 16:28 named.ca  
-rw-r-----. 1 root  named  152 Apr 17 16:28 named.empty  
-rw-r-----. 1 root  named  152 Apr 17 16:28 named.localhost  
-rw-r-----. 1 root  named  168 Apr 17 16:28 named.loopback  
drwxrwx---. 2 named named    6 Apr 17 16:32 slaves
```

Cp the file #cp named.localhost db_forward

```
AAAA      :.1  
[root@server named]# cp named.localhost db_forward  
[root@server named]# ll  
total 20  
drwxrwx---. 2 named named    6 Apr 17 16:32 data  
-rw-r-----. 1 root  root   152 Jun 30 15:50 db_forward  
drwxrwx---. 2 named named    6 Apr 17 16:32 dynamic  
-rw-r-----. 1 root  named 2112 Apr 17 16:28 named.ca  
-rw-r-----. 1 root  named  152 Apr 17 16:28 named.empty  
-rw-r-----. 1 root  named  152 Apr 17 16:28 named.localhost  
-rw-r-----. 1 root  named  168 Apr 17 16:28 named.loopback  
drwxrwx---. 2 named named    6 Apr 17 16:32 slaves
```

Then change root:root to named:named change the owner

```
[root@server named]# chown named:named db_forward
[root@server named]# ll
total 20
drwxrwx---. 2 named named    6 Apr 17 16:32 data
-rw-r-----. 1 named named 152 Jun 30 15:50 db_forward
```

Go to the #vi db_forward

I go to edit the 1. @ ito domainname =shelldecoders. In.

2.aaaa remove it

3.a before add shelldecoders. In. IN

```
[root@server named]# cat db_forward
$TTL 1D
@      IN SOA  shelldecoders.in. rname.invalid. (
                                   0      ; serial
                                   1D     ; refresh
                                   1H     ; retry
                                   1W     ; expire
                                   3H )   ; minimum
      NS   shelldecoders.in.
shelldecoders.in. IN  A      192.168.15.130
```

```
$TTL 1D
@      IN SOA  shelldecoders. in. rname.invalid. (
                                   0      ; serial
                                   1D     ; refresh
                                   1H     ; retry
                                   1W     ; expire
                                   3H )   ; minimum
      NS   shelldecoders. in.
shelldecoders. in. IN  A      192.168.15.130
~
~
~
```

Step 5

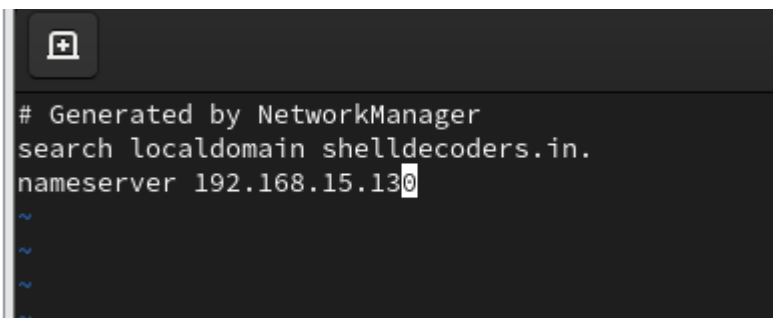
Go 3 configuration file

#vi /etc/resolv.conf

Add shelldecoders. In.

Delete the old ip a and convert the new server ip a

```
[root@server named]# cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain shelldecoders.in.
nameserver 192.168.15.130
```

A terminal window with a dark background. At the top left, there is a small icon of a terminal with a plus sign. The text in the terminal is: # Generated by NetworkManager, search localdomain shelldecoders.in., nameserver 192.168.15.130. There are three blue cursor lines below the text.

```
# Generated by NetworkManager
search localdomain shelldecoders.in.
nameserver 192.168.15.130
```

```
[root@server named]# systemctl restart named.service
```

```
[root@server named]# host shelldecoders.in
shelldecoders.in has address 192.168.15.130
```

```
[root@server named]# nslookup shelldecoders.in
Server:          192.168.15.130
Address:         192.168.15.130#53

Name:   shelldecoders.in
Address: 192.168.15.130
```

